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Group Members: *Tatheer Zahra (084)*

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Course Name: *COAL Project*

Instructor Name: *Sir. Majid Shafique*

Department : *BSCS-3 (B)*

Project Report – Hex Editor Simulation

1. Project Title

Hex Editor Simulation (GUI Based) – COAL Project

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2. Objective

- Understand **memory addressing in computers**
- Read, write, and modify **hexadecimal data**
- Implement **COAL low-level programming concepts** practically

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3. Tools & Technologies

Tool / Technology	Purpose
Visual Studio 2019/2022	C++ GUI development
Win32 API	GUI components (buttons, textboxes)
C++	Programming language
std::ofstream	Memory output file save

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4. Features / Functionalities

1. Memory display (16 bytes: 0x00 – 0x0F)

2. WRITE operation (address + hex value)
3. READ operation (popup with value)
4. CLEAR operation (reset memory)
5. SAVE operation (to memory_output.txt)
6. Error handling (invalid address/value alerts)

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5. Memory Representation (Example)

Address	Hex Value	Description
0x00	00	Default
0x01	3C	User input
0x02	FF	User input
…	…	…
0x0F	0A	Last memory cell

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6. GUI Overview

- Input Boxes: Address & Hex Value
- Buttons: WRITE, READ, CLEAR, SAVE
- Memory Display: Shows all addresses and values

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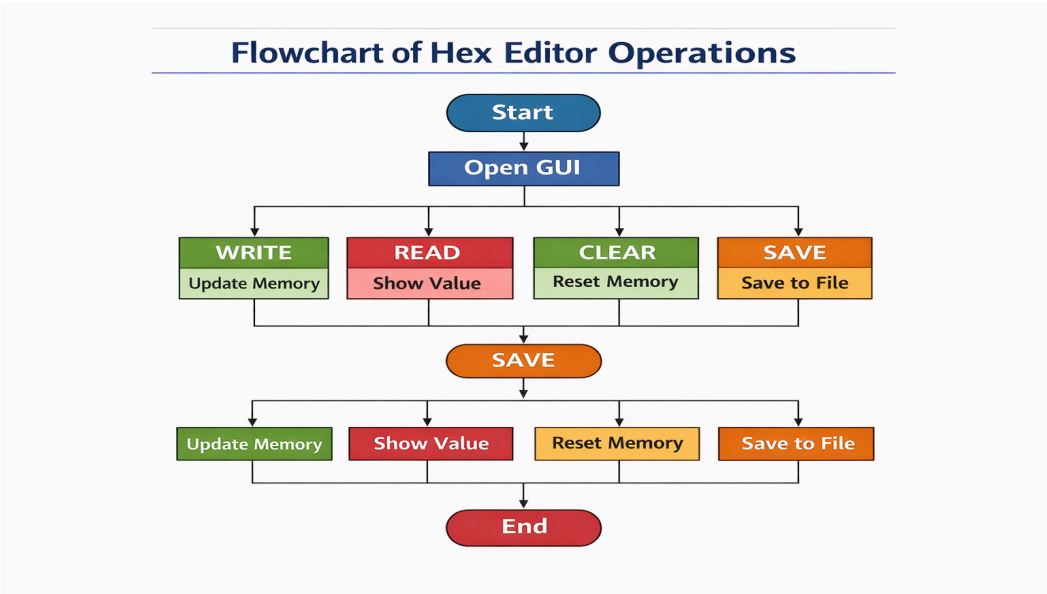
7. Flowchart



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8. Working Steps

1. Run the program
2. Enter address (0–15) and hex value (00–FF)
3. **WRITE** → updates memory
4. **READ** → shows value in popup
5. **CLEAR** → resets memory
6. **SAVE** → stores memory to file

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9. Group Task Distribution

Member Name	Responsibilities
Tatheer Zahra	- Prepare project report (objective, conclusion, key concepts) - Organize GUI screenshots and memory diagram
Tehreem Fayyaz	- Develop GUI-based C++ code - Implement WRITE, READ, CLEAR, SAVE operations - Test memory functionalities and error handling
Malaika Zahoor	- Prepare project document (instructions, screenshots, memory output example) - Integrate file save functionality and format document

This ensures **balanced roles** and smooth viva presentation.

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10. Key Concepts Learned

- Memory addressing
- Hexadecimal number system
- Byte-level memory manipulation
- GUI programming using Win32 API
- File handling for memory persistence

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11. Conclusion

Hex Editor Simulation project demonstrates **address-based memory access, hexadecimal representation, and low-level programming**.
GUI interface with save functionality allows practical understanding of memory operations and simulates real-life hex editor behavior.